APP DEV-1 PROJECT REPORT

PLACEMENT PORTAL APPLICATION

1. STUDENT DETAILS

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ABOUT: I am pursuing B.tech in ECE (3^{rd} year) along with BS in Data Science and Programming. I have keen interest in electronics as well as data driven analysis. I believe in hardworking and multitasking.

2. AI / LLM USAGE

I used **ChatGPT (GPT-5)** to assist in improving the Frontend, guiding in the implementation of browser cache-clearing logic for logout sequence and some error correction.

The overall extent of AI/LLM usage is around 10-15%. All logic building, debugging and implementation were done manually.

3. DESCRIPTION

The Placement Portal Application is a web-based application designed to streamline the campus recruitment process. It acts as a bridge between the college administration, companies and students.

- **Admins** manage the integrity of the system by approving/rejecting companies and drives and by also deactivating existing companies and students.
- **Companies** can post placement drives and manage accept/shortlist/reject student applications.
- **Students** can build profiles, upload resumes, and track their application status in real time.

The system features a robust “Deactivation” logic, allowing admins to deactivate accounts without losing historical data. Also, the admin has the option to completely delete an account from the database.

4. TECHNOLOGIES USED

Technology / Library	Purpose
Flask	The core backend framework logic
Flask-SQLAlchemy	Object Relational Mapper to interact with the database using Python
SQLite	Lightweight relational database to store all records
Jinja2	Templating engine used to inject dynamic data into HTML pages
Bootstrap5	A CSS framework used to create responsive interfaces
HTML5 & CSS3	Standard markup and styling languages to structure and design web pages

5. DATABASE SCHEMA DESIGN

The database design is built around five relational tables:

- **Admin:** A simple table storing the username and password for administrative access.
- **Company:** Stores company details, website, HR contact and status field to manage access control.
- **Student:** Contains student profiles, including name, email, skills and resume.
- **PlacementDrive:** Linked to a specific company via company_id. It stores job specific details like job title, eligibility and deadline.
- **Application:** A bridge table that connects Student and PlacementDrive. It tracks the status of an application (Applied, Shortlisted, Selected or Rejected) and the application date.

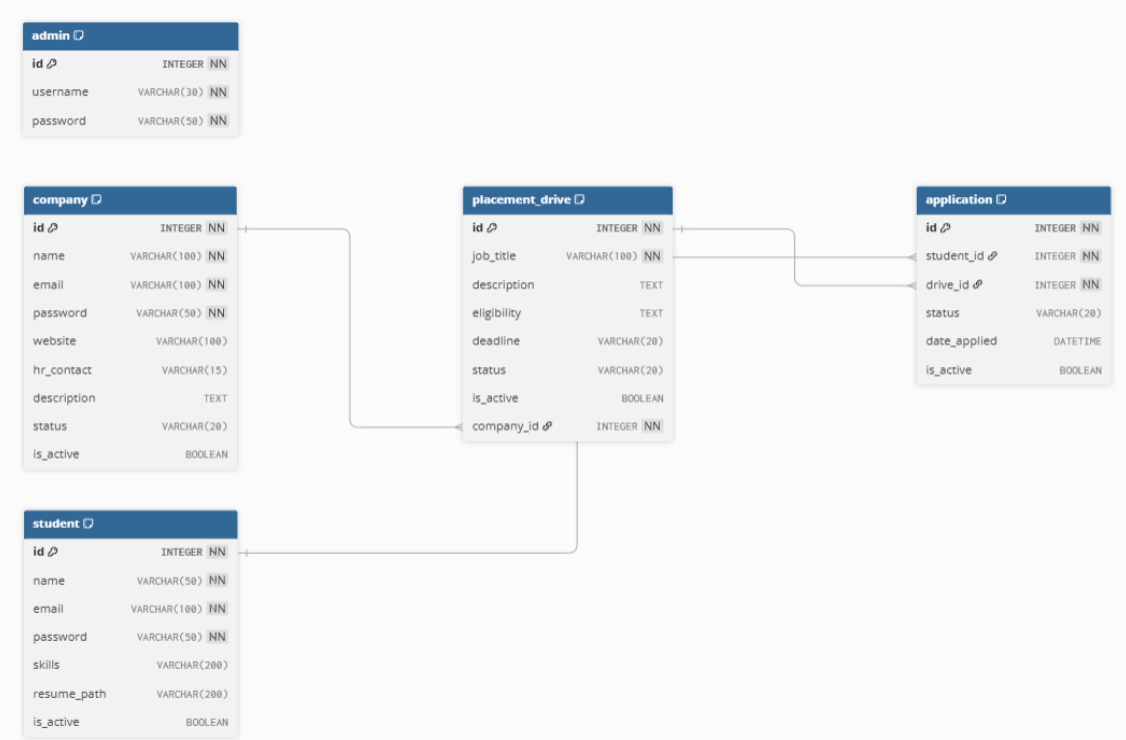

Fig-1: Structure of the Database

Reasons behind the design:

- **Relational integrity:** Using foreign keys ensures that every application is tied to a valid student and a valid job drive.
- **Cascade logic:** Relationships are configured with cascade all and delete-orphan, so that if a company is removed all its drives and applications are automatically cleaned up.

6. ARCHITECTURE AND FEATURES

System Architecture

- **Model:** Managed by models.py, defining the structure and relationships of the data.
- **View:** Handled via HTML and Jinja2 templates, ensuring a clean separation of the user interface from the login.
- **Controller:** Managed in app.py, which processes user input and coordinates between the Model and the View.

Key Features

- **Role Based Access Control:** Session checks prevent students from accessing admin pages and vice-versa.
- **Approval Workflow:** Companies must be approved by the admin before they can login and also any drives posted by the company must be approved by the admin before students can see them.
- **Secure Resume Management:** A standardized naming convention is implemented for uploading student resumes, preventing data overwriting and ensuring easy retrieval.
- **Deactivation logic:** Admin can deactivate students or companies without removing them from the dataset and can re-activate them again if needed.
- **Application integrity:** Logical checks in the backend prevent students from applying to the same drive multiple times.

7. VIDEO PRESENTATION

Drive Link:

https://drive.google.com/file/d/1bHs-_QsjixPI1SBisL3PCXN61714GgUs/view?usp=sharing